

Common discrete distributions

Def. We say that X follows the Bernoulli distribution if

$$P_X(x) = \begin{cases} 1-p & , x = 0 \\ p & , x = 1 \end{cases}.$$

Notation : $X \sim \text{Bern}(p)$, $p \in [0, 1]$.

Interpretation : Conduct an experiment with one trial , in which the prob of success is p . Then X counts the # of success.

$$\text{III} \quad E(X) = (1-p) \cdot 0 + p \cdot 1 = p$$

$$\text{III} \quad E(X^2) = (1-p) \cdot 0^2 + p \cdot 1^2 = p$$

$$\text{III} \quad \text{Var}(X) = p - p^2 = p \cdot (1-p)$$

$$\text{III} \quad M_X(t) = E(e^{Xt}) = e^{0 \cdot t} \cdot (1-p) + e^t \cdot p$$

$$= (1-p) + e^t \cdot p, \quad t \in \mathbb{R}.$$

Def. We say that X follows the binomial distribution if

$$P_X(x) = \binom{n}{x} \cdot p^x \cdot (1-p)^{n-x}, \quad x=0, 1, \dots, n$$

Notation : $X \sim \text{Bin}(n, p)$, $n \in \mathbb{Z}_+$, $p \in [0, 1]$

▮ Interpretation : Conduct an experiment with $\boxed{n}$ number of trials. The prob of success in each trial is $\boxed{p}$. Then X counts the # of successes.

$$\begin{array}{c} \text{▮ } \text{Bin}(n, p) = \text{Bern}(p) \\ \text{||} \\ \text{||} \end{array}$$

▮ Let $Y_i \sim \text{Bern}(p)$, and suppose that Y_i 's are independent. Then

$$X = Y_1 + \dots + Y_n \sim \text{Bin}(n, p).$$

$$\text{▮ } E(X) = E(\cancel{Y_1}^p) + \dots + E(\cancel{Y_n}^p) = n \cdot p$$

$$\text{▮ } \text{Var}(X) = \text{Var}(Y_1) + \dots + \text{Var}(Y_n) = n \cdot p \cdot (1-p)$$

$$\Rightarrow M_X(t) = \prod_{i=1}^n M_{Y_i}(t)$$

$$= \prod_{i=1}^n \left((1-p) + p e^t \right)$$

$$= \left((1-p) + p e^t \right)^n$$

Def. We say that X follows the geometric distribution if

$$P_X(x) = (1-p)^{x-1} \cdot p, \quad x = 1, 2, \dots$$

Notation: $X \sim \text{Geom}(p)$, $p \in [0, 1]$

Interpretation: Keep conducting the same experiment until the first success. The prob of success in each trial is p . X = the # of trials needed.

$$\nabla F_X(x) = P(X \leq x)$$

$$= 1 - P(X > x)$$

$$= 1 - P(\text{the first } x \text{ trials fail})$$

$$= 1 - (1-p)^x$$

$$\nabla M_X(t) = E(e^{xt})$$

$$= \sum_{x=1}^{\infty} e^{xt} \cdot (1-p)^{x-1} \cdot p$$

$$= p \cdot \sum_{x=1}^{\infty} e^{xt - t + t} \cdot (1-p)^{x-1}$$

$$= p \cdot e^t \sum_{x=1}^{\infty} (e^t)^{x-1} \cdot (1-p)^{x-1}$$

$$= p \cdot e^t \sum_{y=0}^{\infty} [e^t \cdot (1-p)]^y$$

$$= p \cdot e^t \cdot \frac{1 - [e^t \cdot (1-p)]^{\infty}}{1 - e^t \cdot (1-p)}$$

$y = x-1$
 $e^t \cdot (1-p) < 1$

$$= \frac{p e^t}{1 - e^t(1-p)} \quad , \quad t < \log\left(\frac{1}{1-p}\right) .$$

$$\text{ii)} \quad E(X) = \frac{d}{dt} M_X(t) \Big|_{t=0}$$

$$= \frac{p e^t(1 - e^t(1-p)) - p e^t \cdot (-e^t(1-p))}{(1 - e^t \cdot (1-p))^2} \Big|_{t=0}$$

$$= \frac{1}{p}$$

$$\text{iii)} \quad E(X^2) = \frac{d^2}{dt^2} M_X(t) \Big|_{t=0}$$

$$= \frac{p + 2 \cdot (1-p)}{p^2}$$

$$\text{iv)} \quad \text{Var}(X) = \frac{p + 2(1-p)}{p^2} - \left(\frac{1}{p}\right)^2 = \frac{1-p}{p^2}$$